Communication Systems Lecture-17: Amplitude Modulation

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Amplitude Modulation (1)

- The complex envelope of an AM signal is given by:

$$g(t) = A_c\,[1 + m(t)]$$

  where $A_c > 0$ and $m(t)$ is the modulating signal.
- Consequently, the AM signal is given by:

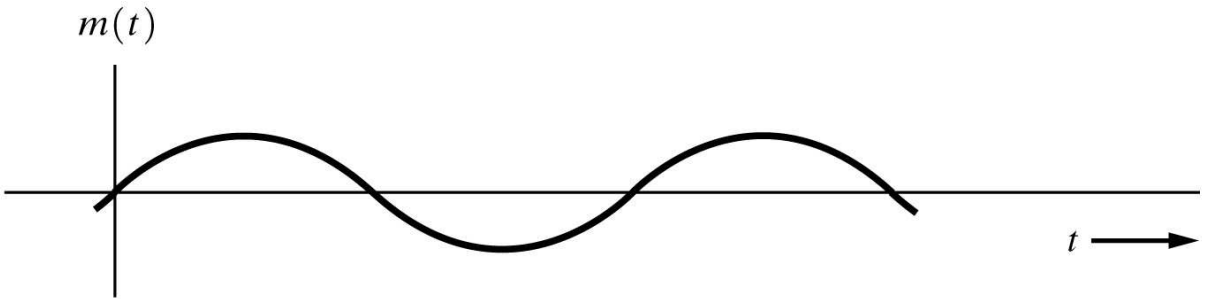
$$s(t) = A_c\,[1 + m(t)]\cos(\omega_c t)$$

  where f_c is the carrier frequency.
- Note that the in-phase component of the complex envelope is $x(t) = A_c\,[1 + m(t)]$ while the quadrature component is $y(t) = 0$.
- The real envelope ($R(t) = |g(t)|$) is equal to $A_c\,[1 + m(t)]$ when $m(t) \geq -1$.
- In what follows, we fix the following notations:

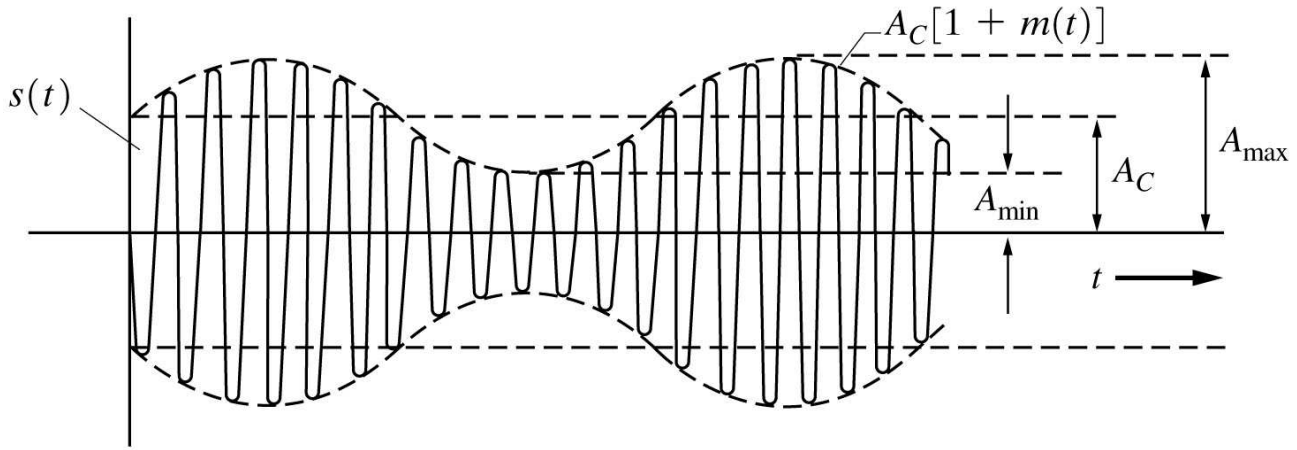
$$A_{min} = \min[g(t)] = A_c\,[1 + \min[m(t)]]$$
$$A_{max} = \max[g(t)] = A_c\,[1 + \max[m(t)]]$$

Amplitude Modulation (2)

- The next figure shows an AM signal when $m(t)$ is sinusoidal (with peak-to-peak value of one). In this figure, $A_{max} = 1.5A_c$ and $A_{min} = 0.5A_c > 0$.



(a) Sinusoidal Modulating Wave



(b) Resulting AM Signal

Amplitude Modulation (3)

- The modulation percentage is defined as:

$$\begin{aligned} \%\ \text{modulation} &= \frac{A_{max} - A_{min}}{2A_c} \times 100 \\ &= \frac{\max[m(t)] - \min[m(t)]}{2} \times 100 \end{aligned}$$

- If $\max[m(t)] = 1$ and $\min[m(t)] = -1$, then the AM signal is said to be 100% modulated.
- When $A_{min} > 0$, the modulating signal $m(t)$ can be recovered by using either:
  - Product detector (with $\theta_0 = 0$).
  - Envelope detector since the real envelope will be the same as the complex envelope: $R(t) = |g(t)| = g(t)$.
- When $A_{min} < 0$, a product detector must be used in order to recover an undistorted version of $m(t)$.
- The product detector is superior to the envelope detector.

Amplitude Modulation (4)

- The average power of the AM signal is:

$$\begin{aligned}\left\langle s^2(t)\right\rangle &= \frac{1}{2}\left\langle g^2(t)\right\rangle = \frac{1}{2}A_c^2\left\langle [1+m(t)]^2\right\rangle \\ &= \frac{1}{2}A_c^2\left\langle 1+2m(t)+m^2(t)\right\rangle \\ &= \frac{1}{2}A_c^2 + A_c^2\left\langle m(t)\right\rangle + \frac{1}{2}A_c^2\left\langle m^2(t)\right\rangle\end{aligned}$$

- Assuming that $m(t)$ contains no DC component, $\langle m(t)\rangle = 0$:

$$\left\langle s^2(t)\right\rangle = \frac{1}{2}A_c^2 + \frac{1}{2}A_c^2\left\langle m^2(t)\right\rangle$$

  - $\frac{1}{2}A_c^2$ is the carrier power.
  - $\frac{1}{2}A_c^2\left\langle m^2(t)\right\rangle$ is the sideband power.

- Note that only the sideband component conveys information.

Amplitude Modulation (5)

- The modulation efficiency is the percentage of the total power that conveys information:

$$E = \frac{\frac{1}{2}A_c^2 \left\langle m^2(t) \right\rangle}{\frac{1}{2}A_c^2 + \frac{1}{2}A_c^2 \left\langle m^2(t) \right\rangle} \times 100 = \frac{\left\langle m^2(t) \right\rangle}{1 + \left\langle m^2(t) \right\rangle} \times 100$$

- For 100% modulation:

$$|m(t)| \leq 1 \quad \Rightarrow \quad m^2(t) \leq 1 \quad \Rightarrow \quad \left\langle m^2(t) \right\rangle \leq 1 \quad \Rightarrow \quad E \leq 50\%$$

The modulation efficiency can not exceed 50% with AM.

- The peak envelope power of the AM signal is:

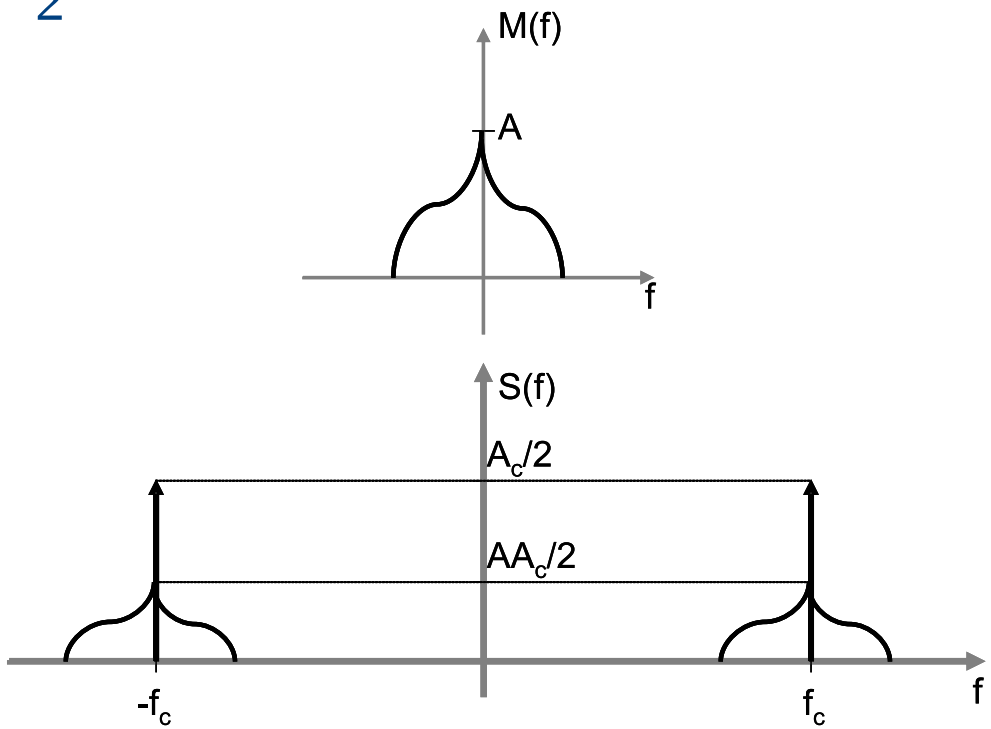
$$\begin{aligned} P_{PEP} &= \frac{1}{2}\left\{\max|g(t)|\right\}^2 \\ &= \frac{1}{2}A_c^2\left\{1 + \max[m(t)]\right\}^2 \end{aligned}$$

Amplitude Modulation (6)

- Given that $s(t) = A_c\left[1 + m(t)\right]\cos(\omega_c t)$, then the spectrum of the AM signal is given by:

$$S(f) = A_c\left[\delta(f) + M(f)\right] * \left[\frac{1}{2}\delta(f - f_c) + \frac{1}{2}\delta(f + f_c)\right]$$

$$= \frac{A_c}{2}\left[\delta(f - f_c) + M(f - f_c) + \delta(f + f_c) + M(f + f_c)\right]$$



- The bandwidth of the AM signal is $2B$ while the bandwidth of the modulating signal is B: AM doubles the bandwidth.

DSB-SC (1)

- A double-sideband suppressed carrier (DSB-SC) signal is an AM signal that has a suppressed discrete carrier. The DSB-SC signal is given by:

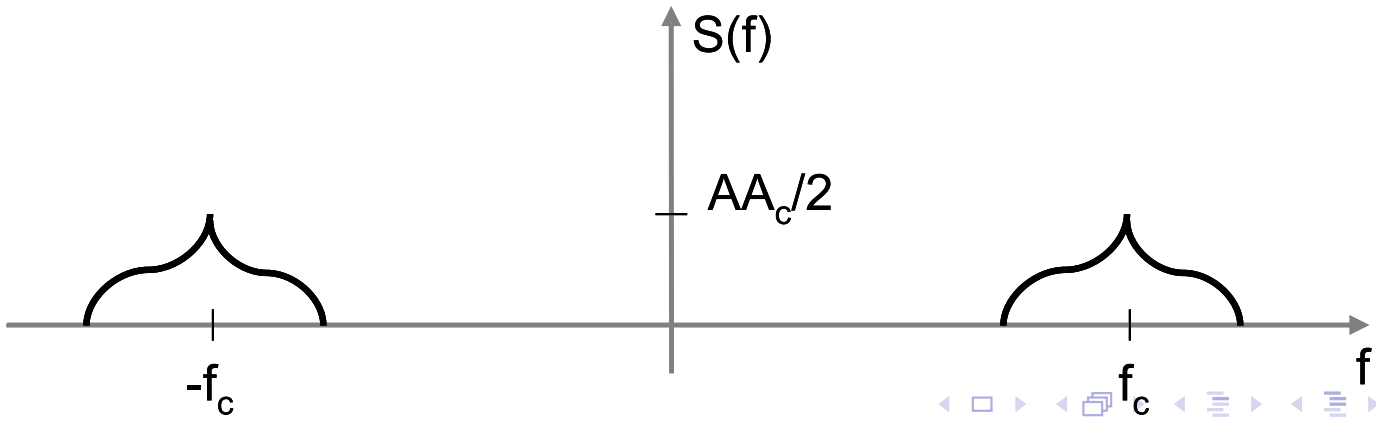
$$s(t) = A_c m(t) \cos(\omega_c t)$$

where $m(t)$ is supposed to have a zero DC level.

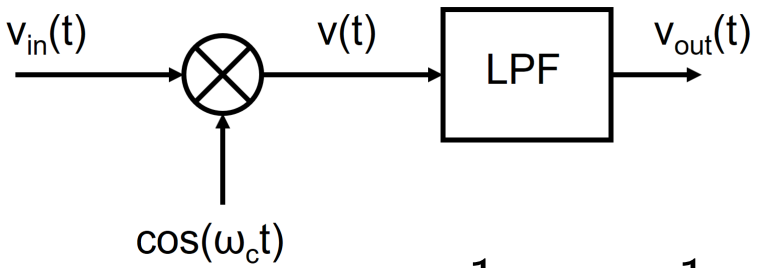
- The spectrum of the DSB-SC signal is:

$$S(f) = \frac{A_c}{2} [M(f - f_c) + M(f + f_c)]$$

- The bandwidth of the DSB-SC signal is double the bandwidth of $m(t)$.



DSB-SC (2)

- The product detector can be used for the demodulation of both AM and DSB-SC signals.
- For both types of modulation, the modulated signal can be written as $v_{in}(t) = \mathrm{Re}\{g(t)e^{j\omega_c t}\} = g(t)\cos(\omega_c t)$ since $g(t) = A_c[1 + m(t)]$ for AM and $g(t) = A_c m(t)$ for DSB-SC where both expressions are real valued.
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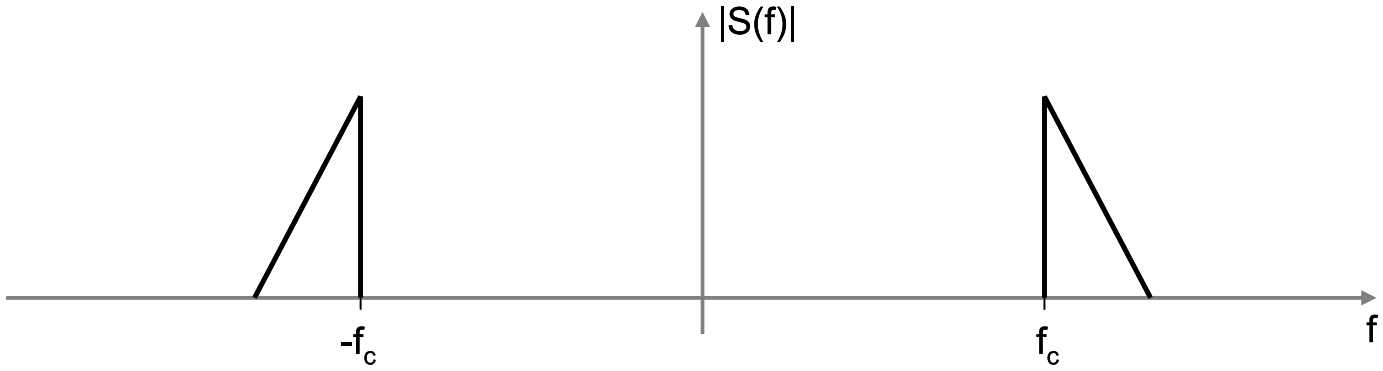
$$v(t) = [g(t)\cos(\omega_c t)]\cos(\omega_c t) = \frac{1}{2}g(t) + \frac{1}{2}\underbrace{g(t)\cos(2\omega_c t)}_{\text{centered around } \pm 2f_c}$$

implying that $v_{out}(t) = \frac{1}{2}g(t)$:

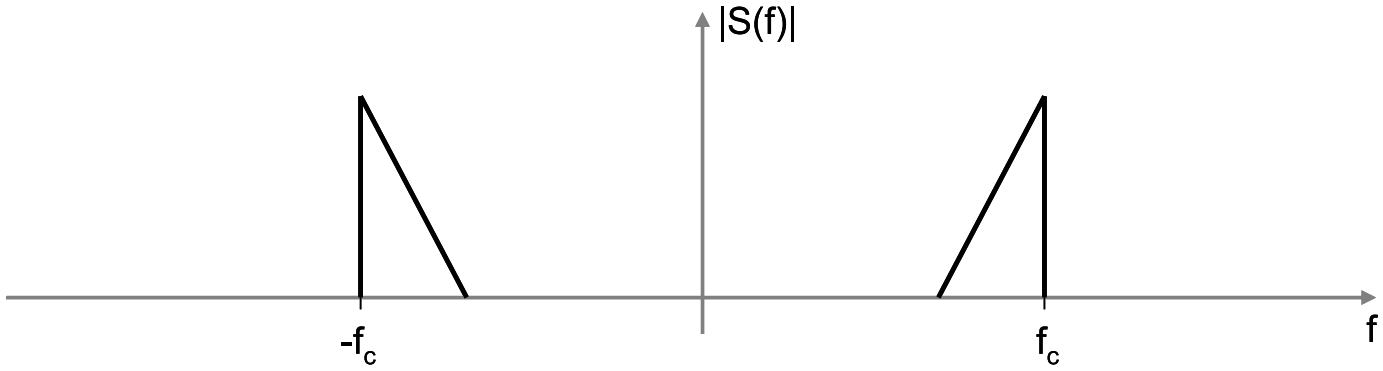
- For AM, $v_{out}(t) = \frac{A_c}{2}[1 + m(t)]$ and the detection is successful.
- For DSB-SC, $v_{out}(t) = \frac{A_c}{2}m(t)$ and the detection is successful.

USSB and LSSB

- An upper single sideband (USSB) signal has a zero-valued spectrum for $|f| < f_c$ where f_c is the carrier frequency.



- A lower single sideband (LSSB) signal has a zero-valued spectrum for $|f| > f_c$ where f_c is the carrier frequency.



USSB-AM (1)

- An USSB-AM signal can be obtained using the complex envelope:

$$g(t) = A_c \left[m(t) + j\hat{m}(t) \right]$$

where $\hat{m}(t)$ denotes the Hilbert transform of $m(t)$ given by:

$$\hat{m}(t) \triangleq m(t) * h(t) \triangleq m(t) * \frac{1}{\pi t}$$

where $H(f) = \mathcal{F}[h(t)]$ corresponds to a -90° phase-shift:

$$H(f) = \begin{cases} -j, & f > 0; \\ j, & f < 0. \end{cases}$$

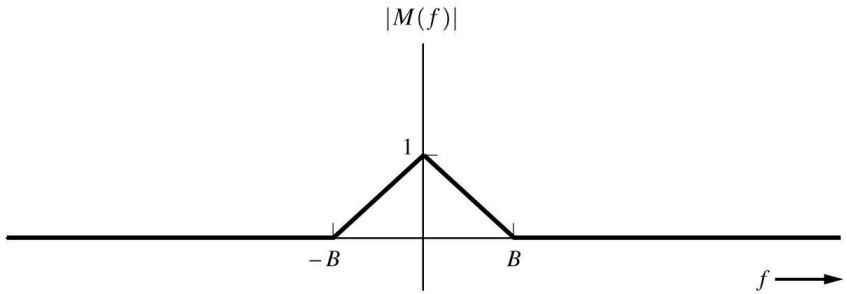
- The Fourier transform of $g(t)$ is:

$$\begin{aligned} G(f) &= A_c \left[M(f) + j\mathcal{F}\left[\hat{m}(t)\right] \right] \\ &= A_c M(f) \left[1 + jH(f) \right] = \begin{cases} 2A_c M(f), & f > 0; \\ 0, & f < 0. \end{cases} \end{aligned}$$

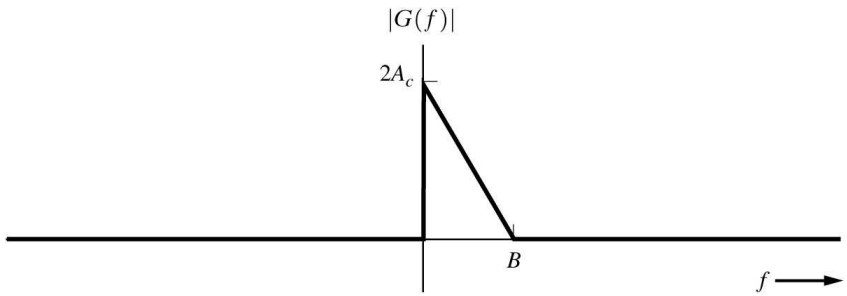
USSB-AM (2)

- The spectrum of the USSB-AM signal is given by:

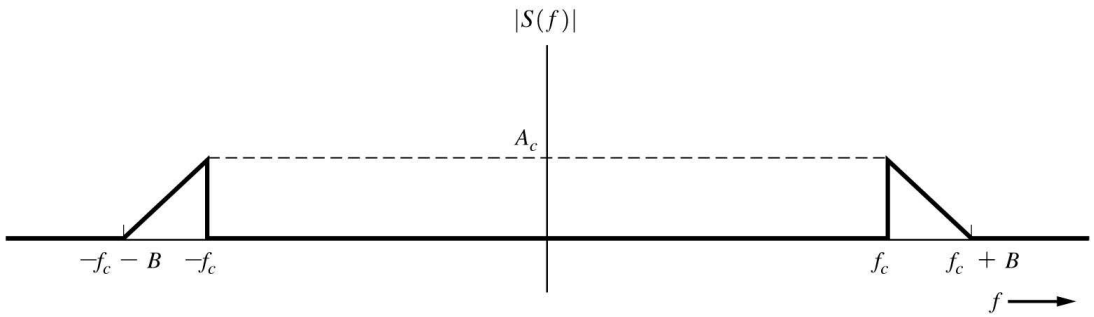
$$S(f) = \frac{1}{2}\left[G(f - f_c) + G^*(-f - f_c)\right]$$



(a) Baseband Magnitude Spectrum



(b) Magnitude of Corresponding Spectrum of the Complex Envelope for USSB



(c) Magnitude of Corresponding Spectrum of the USSB Signal

USSB-AM (3)

- The USSB-AM signal is given by:

$$
\begin{aligned}
s(t) &= \Re\left[g(t)e^{j\omega_c t}\right] \\
&= A_c\Re\left[m(t)e^{j\omega_c t}\right] + A_c\Re\left[\underbrace{j}_{=e^{j\pi/2}}\hat{m}(t)e^{j\omega_c t}\right] \\
&= A_c\Re\left[m(t)e^{j\omega_c t}\right] + A_c\Re\left[\hat{m}(t)e^{j(\omega_c t+\pi/2)}\right] \\
&= A_c m(t)\cos(\omega_c t) + A_c\hat{m}(t)\cos(\omega_c t+\pi/2) \\
&= A_c\left[m(t)\cos(\omega_c t) - \hat{m}(t)\sin(\omega_c t)\right]
\end{aligned}
$$

USSB-AM and LSSB-AM (1)

- In the same way, a LSSB-AM signal can be obtained using the complex envelope:

$$g(t) = A_c \left[m(t) - j\hat{m}(t) \right]$$

- The LSSB-AM signal is given by:

$$s(t) = A_c \left[m(t)\cos(\omega_c t) + \hat{m}(t)\sin(\omega_c t) \right]$$

- Note that USSB-AM signals and LSSB-AM signals occupy the same bandwidth as the modulating signal $m(t)$ (this bandwidth is equal to B).

USSB-AM and LSSB-AM (2)

- The block diagram of a SSB transmitter is shown in the figure below:

